Technology Watch: SWOT Matrix

METHODOLOGICAL FRAMEWORK & WORK GUIDELINES

Conducting a strategic diagnosis in the field of technology watch relies on a rigorous distinction between internal analysis and external analysis:

- **Internal Diagnosis (Strengths / Weaknesses):** Assesses the company's own resources (technological mastery, patents, team skills, financial capacity). These are controllable factors.
- **External Diagnosis (Opportunities / Threats):** Analyzes the scientific, industrial, competitive and regulatory environment. These are factors that are endured or to be exploited.
- **TOWS Cross-Analysis:** Strategic matrix crossing the SWOT: SO (Strengths x Opportunities), ST (Strengths x Threats), WO (Weaknesses x Opportunities), WT (Weaknesses x Threats).
- **Technology Watch Plan (TW):** Continuous system for monitoring scientific, technical, regulatory and competitive information in order to anticipate disruptions.

EXERCISE 01: AgriTech & IoT – Case of the Start-up TechDZ

Context: TechDZ is an innovative Algerian start-up specializing in the design and marketing of IoT sensors for smart agriculture. Its solutions make it possible to optimize irrigation and measure humidity, temperature and luminosity via a cloud platform. The company seeks to strengthen its national positioning while preparing for a regional expansion.

Diagnostic findings:

1. Rapid growth of the connected agriculture market in Algeria.
2. Shortage of IoT skills and specialized staff within the TechDZ team.
3. Reliable, robust products with low energy consumption.
4. Limited marketing and sales budget.
5. Government support and public incentives for AgriTech projects.
6. Strong foreign competition (low-cost Chinese solutions and high-end European ones).
7. Partnerships and collaborations with several local pilot farms.
8. Recurring Internet connectivity problems in rural areas.

Questions:

1.1 Classify each of the 8 findings above in the SWOT matrix (Strength, Weakness, Opportunity, Threat).

1.2 Formulate four (04) cross-referenced operational strategic decisions (SO, ST, WO, WT), justifying your approach.

1.3 Define two (02) major technology watch axes that TechDZ must establish as a priority.

EXERCISE 02: e-Health, Big Data & Sovereignty – Case of the Company HealthDZ

HealthDZ is a young Algerian start-up specializing in the design of connected health solutions. It is developing a digital platform enabling remote medical monitoring through smart medical devices and big data analysis tools.

The goal is to help doctors and patients follow changes in health status in real time, detect anomalies and anticipate certain risks. The connected sensors transmit patients' physiological data to an application that centralizes and analyzes the information using artificial intelligence algorithms. This platform has been tested in a few Algerian hospitals in order to assess its performance and reliability. The project has generated growing interest in the hospital and university community, and the team now aims to extend it to other healthcare facilities.

HealthDZ was founded by a group of engineers in electronics, embedded systems and data analysis. Their mastery of IoT and Big Data technologies enabled them to design an innovative connected health platform adapted to the Algerian context. The company has a young and dynamic team, capable of developing both the sensors and the data processing applications.

The solution has already been field-tested in several pilot hospitals in Algiers, Blida and Constantine, which has provided concrete feedback and made it possible to improve the system's performance. The platform's interface is easy to use, translated into several languages (French, Arabic and English), and designed to make it easy for healthcare professionals to get started.

However, the company faces certain resource limitations. Its funding relies mainly on its own funds and a few national innovation competitions, which limits its capacity to invest in communication, maintenance and the medical certification procedures required to bring healthcare devices to market. The start-up also depends on imported electronic components, whose supply lead times can delay production.

HealthDZ does not yet have a robust IT infrastructure on national territory and relies in part on foreign cloud services, which raises questions of security and sovereignty of medical data. The company's staff has solid technical expertise, but still lacks specific skills in cybersecurity and medical regulation, two essential areas in digital health.

The context in which HealthDZ operates nevertheless offers real growth prospects. The Algerian State actively encourages the digitalization of the healthcare sector and supports innovative local projects. Many international organizations (WHO, European Union, Islamic Development Bank) fund programs to modernize healthcare systems in North Africa, offering opportunities for funding and support.

The e-health market has been expanding rapidly since the COVID-19 pandemic, which revealed the need for remote medical monitoring tools. The Algerian population is increasingly connected to the Internet and equipped with smartphones, which favors the adoption of mobile health applications. Medical institutions are increasingly interested in these technologies, notably to improve care management and prevention.

However, the company must contend with strong foreign competition: several large international brands already offer certified and recognized solutions. These companies have considerable financial resources

and worldwide fame. Moreover, regulation on medical data is becoming increasingly strict. Algerian laws, inspired by the European GDPR, impose high requirements in terms of confidentiality and security. The project also faces infrastructure constraints, as Internet coverage remains uneven across regions. In some rural areas, insufficient connectivity prevents the connected sensors from working properly and limits the reach of the solution. Finally, part of the medical staff remains reluctant to use digital technologies, whether for lack of training or out of fear of replacing the human relationship in care.

Questions:

2.1 Briefly summarize the main objective of the HealthDZ project.

2.2 Carry out a diagnosis of the company HealthDZ (strengths/weaknesses, opportunities/threats).

2.3 Based on your analysis, propose two concrete actions that HealthDZ could implement to strengthen its position in the e-health market

EXERCISE 03: CleanTech & Patent Watch – Case of GreenEnergyDZ

GreenEnergyDZ is a technology company (CleanTech) based in the heart of the Hassi R'Mel energy hub (Algeria). It specializes in the design and manufacture of smart hybrid energy management modules (EMS – Energy Management System), capable of optimizing in real time the coupling between photovoltaic power plants and green hydrogen production units (electrolyzers).

The company relies on a small but highly qualified technical team that has developed unique algorithmic know-how in the predictive optimization of Solar/Hydrogen energy yield. Its local presence in Hassi R'Mel gives it direct access to national energy infrastructure and an ideal testing ground under strong solar potential. Thanks to its agile structure, GreenEnergyDZ is able to evolve its software very quickly.

However, the company suffers from a very limited R&D budget compared with the global energy giants. It holds no internationally filed patent to date to protect its innovations, and its profitability is hit hard by its heavy dependence on imports of electronic components and rare metals (semiconductors, electrodes) needed to assemble its units.

On the external front, GreenEnergyDZ operates in a very favorable context: the Algerian government has launched a National Plan for the Development of Green Hydrogen, offering funding, subsidies and an incentive regulatory framework. In addition, strong demand from the European Union to import clean energy represents a considerable potential export market. Moreover, many technologies patented abroad have not been filed with the INAPI (Algerian National Institute of Industrial Property), which makes them legally exploitable on national territory.

On the other hand, the sector faces a proliferation of blocking patents filed aggressively by foreign multinationals, which exposes GreenEnergyDZ to the risk of infringement lawsuits if it tries to export. To this are added the extreme volatility of world prices of raw materials (iridium, platinum, lithium) and the risk of reverse engineering of its units by international competitors with greater financial means.

Questions:

3.1 Draw up the SWOT matrix of GreenEnergyDZ.

3.2. Demonstrate how Patent Watch / Industrial Property monitoring helps avoid infringement and identify public-domain technologies.

3.3. Draft 3 key recommendations for the R&D Director.